

AYUSH AGGARWAL, PHD

COMPUTATIONAL BIOLOGIST

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PROFESSIONAL SUMMARY

PhD scientist with 10+ years of experience translating high-dimensional, multimodal genomics data into actionable biological insights. Analyzed >500k cells across 50+ samples, contributing to 11 peer reviewed publications. Expertise in developing reproducible computational pipelines for single-cell, spatial, and bulk omics data, with strong proficiency in R, Python, and HPC. Complementary wet-lab expertise spanning NGS library preparation and human biospecimen processing. Looking for Senior Data Scientist, Senior Bioinformatician or similar position.

PROFESSIONAL EXPERIENCE

POSTDOCTORAL RESEARCHER | UNIVERSITY OF CALIFORNIA, SAN FRANCISCO

JAN 2024 – PRESENT

Cancer neuroscience research integrating multimodal genomics data to study meningioma biology.

- **Meningioma Brain Invasion:** Integrated bulk transcriptomics from 199 tumors with patient-matched single-cell and spatial transcriptomics to resolve molecular and cellular changes at the brain-tumor interface; identified conserved tumor reprogramming and microenvironment interactions associated with brain invasion (**Neuro-Oncology, 2025**).
- **Meningioma Sex Specificity:** Bioinformatics support to analyze transcriptomics, proteomics, and epigenomics data, identifying a 58 gene biomarker of Progesterone activity leading to the discovery of RNA-processing mechanisms underlying sex-biased tumor biology (**Manuscript Under Review at *Nature Portfolio***).
- **Meningioma therapeutic vulnerabilities:** Leveraging Perturb-seq and scRNA+ATAC data to identify candidate drug targets (**Manuscript under preparation**).
- **Meningioma intra-tumor heterogeneity:** Analysis of spatial transcriptomics, DNA methylation and RNA-seq data to study meningioma intra-tumor heterogeneity

GRADUATE STUDENT | CSIR-IGIB, INDIA

AUG 2017 – DEC 2023

Skin pigmentation research leveraging computational and experimental approaches to study melanocyte cell state heterogeneity and its implications in pathological conditions.

- **Multimodal Genomics:** Investigated the molecular basis of melanocyte cell-state heterogeneity by integrating scRNA-seq, RNA+ATAC, and ChIP-seq with high content imaging and flow cytometry, uncovering transcriptional and epigenetic programs governing proliferative and mature states, subsequently developed MelDat, an R/Shiny application for interactive exploration and visualization of these datasets (**PLOS Biology, 2024**).
- **Candidate Discovery:** Integrated multiple single-cell and bulk transcriptomic datasets to identify candidate regulators of melanocyte development and function, resulting in the discovery of MGAT4B as a key regulator of melanocyte development (**PNAS, 2025**).
- **Assay Development:** Developed and validated a low-effort imaging flow cytometry assay for quantitative melanin estimation at single-cell resolution, enabling characterization of heterogeneous pigmentation states without conventional melanin extraction.
- Multiple inter and intra-Institution collaborations resulting in 7 publications.

RESEARCH ASSISTANT | CSIR-IGIB, INDIA

OCT 2015 – JUL 2017

Investigated transcriptional regulation of *Mycobacterium* fatty-acid metabolism using molecular cloning and GFP reporter assays under various growth conditions including biofilm.

TECHNICAL SKILLS

- **PROGRAMMING & COMPUTING:** R | Python | Bash | HPC | Git | ImageJ scripting
- **GENOMICS & BIOINFORMATICS:** Single-cell & Spatial Omics | DNA Methylation | Bulk Transcriptomics | Epigenomics | Proteomics | Seurat | Scanpy | Targets | Nextflow
- **MACHINE LEARNING:** tidymodels | Regression | Random Forest | Model Training & Cross-Validation
- **EXPERIMENTAL:** NGS Library Preparation | Human Biospecimen Processing | Mammalian Cell Culture | (Imaging) Flow Cytometry | High-Content Imaging | Molecular Cloning | qPCR | Immuno-cytochemistry

EDUCATION

PHD, SCIENCE | CSIR-INSTITUTE OF GENOMICS AND INTEGRATIVE BIOLOGY, INDIA
B.E., BIOTECHNOLOGY | DELHI UNIVERSITY, INDIA

2017 – 2023
2011 – 2015